4.2 Differentials, Linear Approximations, and Newton's Method

Name _____ Period ____ Date _____

AP Calculus 4.2 · Student sheet

I can…

- find a linear approximation and use it to estimate a value.
- find the differential dy and use it to estimate error.
- use Newton's Method to approximate a zero.

Vocabulary · fill in the missing word

Linear approximation — $L(x) = f(x_0) + f'(x_0)(x - x_0)$, the tangent line used to _____ f near x_0

Differential — dy = _____

Relative error — the error divided by the _____ value, about dy/y

Newton's Method — x_{n+1} = _____

Part 1 · Zoom on the tangent

Compare $y = \sqrt{x}$ with its tangent line $L(x) = (1/4)x + 1$.

x	√x	L(x)	Difference
4.1			
4.5			
5			

Part 2 · Notes and examples (fill in each note, then work the example)

Linear approximation: $L(x) = f(x_0) +$ _____$(x - x_0)$.

Example 1

Find the linear approximation of $f(x) = \sqrt{x}$ at $x = 4$, and use it to approximate $\sqrt{4.1}$.

Example 2

Use a linear approximation to approximate $\sqrt{26}$.

Differentials: dy = _____ dx, and $\Delta y \approx dy$ when dx is small.

Example 3

$y = x^3$. Find dy and Δy when $x = 2$ and $dx = \Delta x = 0.1$.

Relative error: if $V = kx^3$, then $dV/V =$ _____ $\cdot\, dx/x$.

Example 4

The radius of a sphere is measured with a possible error of ±1%. Estimate the percentage error in the volume.

Newton's Method: $x_{n+1} = x_n -$ _____.

Example 5

Use Newton's Method with $f(x) = x^2 - 5$ and $x_1 = 2$ to find x_2.

Part 3 · Partner practice

Partner 1

Use a linear approximation to approximate $\sqrt{65}$.

Partner 2

(AP® Practice) The linear approximation of $f(x) = e^{2x}$ at $x = 0$ is used to approximate $f(0.1)$. The approximation is

(A) 1.2 (B) 1.1 (C) 1.22 (D) 0.8